Conditional Statements

simple if

In [1]:
```
age = int(input("enter your age:"))
if age > 18 :
    print("Elligible for voting.")
```

if_else

In [3]:
```
age = int(input("enter your age:"))
if age > 18 :
    print("Elligible for voting.")
else:
    print("Can not vote.")
```

```
Elligible for voting.
```

ternary operator

In [5]:
```
x, y = 10, 20
res = x if x > y else y
print("Largest numb`er is :", res)
```

```
Largest numb`er is : 20
```

nested if else

In [8]:
```
age = 65
isMember = False
if age > 60:
    if isMember:
        print("30%")
    else:
        print("20%")
else:
    print("No discount")
```

```
20%
```

Largest among three

In [11]:
```
x, y, z = 45, 25, 96
if x > y and x > z:
    res = x
elif y > z:
    res = y
else:
    res = z
print(res)
```

```
96
```

In [13]:
```
if x > y and x > z:
    res = x
else:
    if y > z:
        res = y
    else:
        res = z
```

In [15]:
```
x if x > y and x > z else y if y > z else z
```

Out[15]: 96

match case

In [18]:
```
x = 100
match x:
    case 10:
        print("10")
    case 20:
        print("20")
    case 30:
        print("30")
    case _:
        print("This is default case") #wildcard case
```

```
This is default case
```

In [20]:
```
x = 10
match x:
    case 30 | 40 | 50 | 60:
        print("30")
    case 10 if x%2==0:
        print("10 with condition")
    case 10:
        print("10")
    case _:
        print("This is default case")#wildcard case
```

```
10 with condition
```

Looping Statements : for & While

while loop

In [24]:
```
i = 1
while i < 6:
    print(i, end = " ")
    i += 1
```

```
1 2 3 4 5
```

In [26]:
```
i = 10
while i >= 1:
    print(i, end = " ")
    i -= 2
```

```
10 8 6 4 2
```

continue

In [29]:
```
i = 0
while i < 5:
    i += 1
    if i == 3:
        continue
    print(i)
```

```
1
2
4
5
```

break

In [32]:
```
i = 0
while i < 5:
    i += 1
    if i == 3:
        break
    print(i)
```

```
1
2
```

while...else

In [35]:
```
i = 1
while i <= 5:
    if i == 3:
        break
    print(i)
    i += 1
else:
    print("Loop terminated successfully.")
print("End of loop.")
```

```
1
2
End of loop.
```

In [37]:
```
i = 1
while i <= 5:
    if i == 30:
        break
    print(i)
    i += 1
else:
    print("Loop terminated successfully.")
print("End of loop.")
```

```
1
2
3
4
5
Loop terminated successfully.
End of loop.
```

for loop

In [40]:
```
for i in range(5):
    print(i, end = " ")
```

```
0 1 2 3 4
```

In [42]:
```
for i in range(1,5):
    print(i, end = " ")
```

```
1 2 3 4
```

In [44]:
```
for i in range(1,10, 2):
    print(i, end = " ")
```

```
1 3 5 7 9
```

In [46]:
```
for i in range(-10, 0, 1):
    print(i, end = " ")
```

```
-10 -9 -8 -7 -6 -5 -4 -3 -2 -1
```

In [48]:
```
for i in range(-10, 0, 2):
    print(i, end = " ")
```

```
-10 -8 -6 -4 -2
```

In [50]:
```
for i in range(10, 0, -3):
    print(i, end = " ")
```

```
10 7 4 1
```

In [52]:
```
s1 = "Python"
for i in s1:
    print(i)
```

```
P
y
t
h
o
n
```

In [54]:
```
li = [10, 20, 30, 40]
for i in li:
    print(i)
```

```
10
20
30
40
```

for else

In [57]:
```
for i in range(1,10):
    if i%7 == 0:
        break
    print(i, end = " ")
else:
    print("\nLoop terminated successfully.")
print("\nEnd of loop.")
```

```
1 2 3 4 5 6
End of loop.
```

In [59]:
```
for i in range(1, 10):
    if i%20 == 0:
        break
    print(i, end = " ")
else:
    print("\nLoop terminated successfully.")
print("End of loop.")
```

```
1 2 3 4 5 6 7 8 9
Loop terminated successfully.
End of loop.
```

Examples

Split a number

In [63]:
```
n = 1234
while n > 0:
    r = n % 10
    print(r)
    n = n // 10
```

```
4
3
2
1
```

count total number of digits in a number

In [66]:
```
n = 1234
dig = 0
while n > 0:
    dig += 1
    n = n // 10
print("Total no. of digits in",n,"is/are",dig)
```

```
Total no. of digits in 0 is/are 4
```

reverse a number

In [69]:
```
n = 1234
rev = 0
m = n
while n > 0:
    r = n % 10
    rev = rev * 10 + r
```

```
    n = n // 10
print("Reverse of",m,"is",rev)
```

```
Reverse of 1234 is 4321
```

check whether a number is Palindrome or not

In [72]:
```
n = 1111
rev = 0
m = n
while n > 0:
    r = n % 10
    rev = rev * 10 + r
    n = n // 10
if m == rev:
    print(m,"is Palindrome.")
else:
    print(m,"is not Palindrome.")
```

```
1111 is Palindrome.
```

Find the factors of a number

In [75]:
```
n = 12
print("Factors of",n,"are :")
for i in range(1,n+1):
    if n%i == 0:
        print(i)
```

```
Factors of 12 are :
1
2
3
4
6
12
```

Find whether a number is Prime or not - using counter variable.

In [78]:
```
n = 7
count = 0
for i in range(1,n+1):
    if n%i == 0:
        count += 1
if count == 2:
    print(n,"is a Prime number.")
else:
    print(n,"is not a Prime number.")
```

```
7 is a Prime number.
```

Find whether a number is Prime or not - using for_else

In [81]:
```
n = 7
for i in range(2,n):
    if n%i == 0:
        print(n,"is not a Prime number.")
```

```
        break
else:
    print(n,"is a Prime number.")
```

```
7 is a Prime number.
```